



EVOLVING FEniCS: THE EXTENSION ECOSYSTEM AT SIMULA SCIENTIFIC COMPUTING

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The FEniCS project provides the foundational algorithms for finite element computations, which in turn has organically cultivated a rich ecosystem of software extensions. This talk will highlight several key extensions developed and maintained by Simula Research Laboratory. Specifically, I will introduce SciFem, a utility toolbox for DOLFINx; io4dolfinx, a modern IO-library; and FEniCSx_ii, a library designed for non-local operators. The presentation will conclude with a look ahead at frameworks currently under active prototyping, including DXA (dolfinx-adjoint), FEniCSx-Jax and networks_fenicsx.